

A Formally Verified Library of Mathematical Finance in Lean 4

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Abstract

We describe a library of mathematical finance in the Lean 4 proof assistant, built on Mathlib and the `BrownianMotion` package. Its benchmark records 535 results from the literature, each with a Lean statement, a status, and a published account of what the formalization leaves out: 504 are proved in full, 18 restate an upstream theorem, and 13 are reduced cores that prove less than the result they are named after. The library is about 100,000 lines of Lean, contains no `sorry`, and a build-time audit pins every library constant that a benchmark proof cites to Lean’s three standard axioms. It reaches from continuous-time foundations, where the Itô integral is constructed rather than assumed and carries Itô’s formula without growth conditions, martingale representation, Girsanov’s theorem for bounded predictable drifts and Lipschitz stochastic differential equations, through no-arbitrage theory and derivative pricing to quantitative risk management, fixed income, portfolio theory and actuarial mathematics. Two things make it more than a catalogue. One is depth: the objects of continuous-time pricing are constructed rather than posited, from Brownian motion and an Itô integral built as an isometry. The other is an audit of faithfulness that makes the logical status of every claim checkable, and whose review has caught what mechanical gates cannot see, including a convergence theorem whose hypothesis could never be satisfied. Nearly all of the mathematics is classical, and what a formal base over it produces is certified unification of known results. The library also hosts two contributed developments, by Robert Martin, on the American put’s exercise boundary and on Durrleman’s condition for SVI smiles, that address problems its own survey had listed as open. The contribution is methodological and infrastructural.

1 INTRODUCTION

Interactive theorem provers have moved, over the last decade, from isolated landmark proofs to broad, reusable libraries. Mathlib [18] now carries a substantial measure-theoretic probability development: conditional expectation, martingales, the Radon–Nikodym theorem, Doob’s theorems, Gaussian measures, and now the definition of pre-Brownian motion. Degenne and collaborators built Brownian motion itself on top of it, in the `BrownianMotion` package [8, 4]. Mathematical finance sits directly on this foundation: arbitrage-free pricing, the Black–Scholes model [3], the fundamental theorem of asset pricing [13], risk measurement. Little of it had been formalized.

What existed was narrow and deliberate. Echenim and Peltier formalized the binomial model and its completeness in Isabelle/HOL [10], and with Guiol the Cox–Ross–Rubinstein pricing of European derivatives [11]. Pusceddu and Bartoletti formalized

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constant-product automated market makers in Lean 4 [21]. Nagy gave a machine-verified derivation from Itô’s lemma to Black–Scholes in Lean 4 [20], a focused vertical that, on its author’s account, leaves the continuous Itô integral structurally verified rather than constructed. More recently, Ushakov and Berdinsky priced European and barrier options through a Henstock–Kurzweil path integral in Lean [25]. Each of these is self-contained, and none offers a way to state *how faithful* a formalization is to the mathematics it claims.

This paper describes a library that is broad, deep in its continuous-time foundations, and audited for faithfulness. It covers thirteen areas, from the Itô integral to quantitative risk management, and the state reported here is release v1.5.0 of October 2026. Its Itô calculus and its fundamental theorems of asset pricing are treated in detail in two companion papers [6, 5]; this one describes the library as a whole, the principles that keep it coherent, and the audit that keeps it honest.

Changes from the third version. The corpus has grown from 330 to 535 entries and the library has nearly doubled, from about 52,000 to about 100,000 lines. New since the third version are martingale representation and market completeness, Itô’s formula for Itô processes, the agreement of the integral with the `BrownianMotion` package’s stochastic integral, jump-diffusions, a quantitative-risk-management benchmark of 65 entries, and two contributed developments (Section 9); some results the library already contained, among them Itô’s formula under exponential growth and stochastic differential equations, are reported here for the first time. The audit now pins every cited constant rather than a curated list. Several statements of the third version are corrected here: the gloss of the reduced-core tier (Section 7), the list of instances of the Garman normal form and the account of the survival identity (Section 6), the status of the binomial limit, whose stated hypothesis was unsatisfiable at the time (Section 7.3), and the description of the integral $\int B \, dB$ (Section 5).

2 CONTRIBUTIONS

- (1) *Breadth on one foundation* (Section 4). To our knowledge the broadest formalization of mathematical finance in any proof assistant: thirteen areas and 535 catalogued results on one set of definitions, with cross-module reuse rather than per-instrument re-derivation.
- (2) *Depth in continuous time* (Section 5). The Itô integral is constructed as an isometry, and Itô’s formula, martingale representation, Girsanov’s theorem and stochastic differential equations are built on it; the risk-neutral measure is constructed by a change of measure rather than assumed, and the discounted price is identified as an Itô integral with its integrand named.
- (3) *A faithfulness audit* (Section 7), the most transferable contribution. Every result carries a tier that records how its Lean statement relates to the claim, and a published disclosure of what it leaves out. Build-enforced gates pin axioms, forbid unproved or definitional shortcuts in full entries, keep verification evidence fresh, and stop prose from naming an object a statement only asserts to exist. Section 7.3 reports what the gates missed and how it was found.
- (4) *An honest map* (Sections 8 and 10). The reduced cores, the limitations of the headline results and of the apparatus itself, and the provenance of the code, including third-party ports and AI assistance (Section 9).

	this work	Nagy (2026)	Echenim et al. (2017–20)
prover	Lean 4	Lean 4	Isabelle/HOL
scope	13 areas, 535 results	one derivation (Itô to Black–Scholes)	binomial and CRR models, completeness
continuous Itô integral	constructed	structurally verified	not applicable (discrete time)
risk-neutral measure	derived by a change of measure	derived in a two-state market	discrete-time model
faithfulness audit	four tiers, per-entry disclosure, axiom gates	not part of the work	not part of the work

TABLE 1. This work against the nearest formalized-finance precedents.

3 RELATED WORK AND POSITIONING

Formalized finance has appeared mostly at formal-methods venues. [Table 1](#) compares the nearest precedents along the two axes the prior art leaves open. Nagy’s derivation [20] is the closest, and the library builds on it directly: three modules (`Foundations/DiscreteIto.lean`, `FTAPTTwoState.lean`, `ItoLemma.lean`) adapt its discrete Itô, two-state fundamental theorem and structural-drift framework, with attribution in their headers. Where Nagy machine-checks one Itô-to-Black–Scholes path and stops at the continuous Itô integral, this library constructs that integral and sets the same derivation inside a comprehensive, audited development. The work of Echenim, Guiol and Peltier [10, 11] is the established precedent for formalized pricing in discrete time. The library re-implements the constant-product invariants of Pusceddu and Bartoletti [21] for breadth, adapting their mathematical content with attribution, and claims nothing new in automated market makers. Ushakov and Berdinsky [25] take a different analytic route to pricing, through the Henstock–Kurzweil path integral rather than a stochastic integral.

Mathlib and the `BrownianMotion` package are the upstream foundation. The library consumes Mathlib’s measure theory, conditional expectation, martingales and Gaussian machinery, and the package’s construction of Brownian motion, its simple processes and predictable σ -algebra, local martingales and axiomatic stochastic integral; [Section 8](#) reports what has been contributed back. On the benchmark side, `StochBench` [24] collects 450 graduate stochastic-process problems in Lean and, lacking infrastructure such as the Itô integral, states 336 of them with the required properties as hypotheses. The library’s benchmark is of a different kind: it records results that are proved, and its reduced-core tier makes exactly that hypothesis-taking visible where it remains.

4 WHAT THE LIBRARY PROVES

[Tables 2](#) and [3](#) list the thirteen areas with representative results. Each one-line statement describes the Lean statement, not the textbook theorem it is named after; where they differ, the difference is given.

No individual theorem in the table is new; the mathematics is classical throughout, with the exception of the two contributed developments discussed in [Section 9](#). The contribution is the development as a whole: one foundation under all of it, depth where earlier efforts assume, and an audited statement of what each result proves.

Area	Representative results
Probability	Doob's L^p maximal inequality for non-negative submartingales (<code>maximal_ineq_Lp</code>); convergence of right-continuous L^p -bounded martingales, $p > 1$, almost surely along the integers and in measure along the reals; the Ionescu-Tulcea path measure of a Markov chain on cylinders.
Stochastic calculus	The Itô integral as an isometry and as a continuous local martingale; Itô's formula in L^2 under bounded or exponentially growing derivatives, without growth conditions as a local martingale (<code>ito_formula_unrestricted</code>), and for Itô processes with bounded adapted continuous coefficients (<code>ito_formula_adapted</code>); martingale representation; Girsanov's theorem for bounded predictable drifts, at the level of finite-dimensional laws; Lipschitz SDEs on short horizons; the heat-equation link of Feynman-Kac [6].
No-arbitrage	The fundamental theorem over several periods on a finite probability space with one risky asset (<code>ftap_discrete</code>), and in one period with finitely many assets, on a finite state space and on any probability space (<code>ftap_one_period_vector</code>) [5]; an equivalent martingale measure excludes arbitrage by simple strategies in continuous time; change of numéraire; completeness of the Brownian market with a unique hedge (<code>exists_replicating_strategy</code>); uniqueness of the pricing measure under a square-integrable density.
Option pricing	The Black-Scholes call under a lognormal hypothesis that is itself derived from the physical model; the Black-Scholes equation from Feynman-Kac (<code>bsV_satisfies_bs_pde_via_feynmanKac</code>); Breeden-Litzenberger in second-order form for Black-Scholes and for jump-diffusions with a Gaussian part, and in first-order form (call prices at all strikes determine the law) for any law with a finite forward; jump-diffusions, assumed as a hypothesis structure: the discounted price is a martingale exactly at the compensated drift (<code>JumpDiffusionProcess.martingale_iff</code>), Merton's 1976 series at every date before maturity (<code>condExp_call_eq_mertonCallPrice</code>), Esscher pricing, and incompleteness at one date (<code>exists_call_esscher_ne_merton</code>).
Exotic options	The n -date geometric Asian call in Black-Scholes form; Margrabe's formula as a Black-Scholes call on the price ratio when the ratio is lognormal, and the change-of-numéraire identity for the exchange payoff; quanto forwards; variance swaps, with realized variance converging to $\sigma^2 T$ in L^2 for any drift.
Lattice models	The Cox-Ross-Rubinstein call price converges to the Black-Scholes price (<code>binomialPrice_call_tendsto_bs_closed</code>), with a lemma that its no-arbitrage hypothesis holds when $ r \sqrt{T} < \sigma$ (<code>binomialNoArb_crr</code>); American options as Snell envelopes; André's reflection principle as a bijection of lattice paths.

TABLE 2. Representative results in probability, stochastic calculus and pricing. Every name is a Lean declaration in the library.

Area	Representative results
Fixed income, credit	The Vasicek bond price in affine form, with the integrated rate defined through its Wiener representation; the forward measure as a change of numéraire, in a constant-rate model where it coincides with the risk-neutral measure; duration and convexity as price derivatives; hazard rates as logarithmic derivatives of survival; the CDS fair spread from an assumed balance of premium and protection legs; under independence, the first-to-default hazard as the sum of the single-name hazards; the KMV–Merton default probability through the distance to default.
Portfolio theory	CAPM from the equilibrium first-order condition; the Black–Litterman posterior mean as the unique solution of its precision-weighted normal equation, without the Bayesian derivation; Markowitz critical points from the Lagrangian; Kelly growth over n periods; the Kelly portfolio as numéraire induces the risk-neutral measure in a two-outcome market.
Risk measures	The Artzner–Delbaen–Eber–Heath representation of coherent risk measures on a finite state space (<code>coherentRisk_isLUB</code>); value-at-risk as the quantile of an arbitrary law; the Rockafellar–Uryasev formula, coherence and dual representation of expected shortfall for every integrable loss; elicibility of value-at-risk, on laws with a finite mean and a unique quantile, and its failure for expected shortfall.
Dependence, extremes	Sklar’s theorem for continuous margins; Fréchet–Hoeffding bounds; max-stability of the GEV law; the generalized Pareto law in its domain of attraction; maxima of Poisson-many losses. Fisher–Tippett and Pickands–Balkema–de Haan are not formalized.
Credit portfolios	The value-at-risk of the one-factor Gaussian default fraction converges to the asymptotic single-risk-factor quantile, the worst-case default rate at the core of the Basel IRB formula (<code>tendsto_valueAtRisk_oneFactorLossFraction</code>).
Microstructure	A positive Glosten–Milgrom spread from trader-mix primitives; for one asset, the quadratic ansatz of Bergault–Evangelista–Guéant–Vieira solves the approximate (linear–quadratic) Hamilton–Jacobi equation whenever its coefficients solve the associated ordinary differential equations, and for several assets the matrix Riccati equation of the leading coefficient is verified; the true value function is out of scope.
Other	A contract language that separates payoffs from pricing models, after Bilokon [2]; survival models, annuity-certain closed forms and compound Poisson losses; constant-product AMMs; strict convexity of the American put’s exercise boundary and a polynomial-sign characterisation of Durrleman’s condition for raw SVI, both by Robert Martin, the latter extended by the library to butterfly-free smiles (Section 9).

TABLE 3. Representative results in the applied areas. Every name is a Lean declaration in the library.

5 DEPTH: THE CONTINUOUS-TIME SPINE

The library’s depth comes from carrying stochastic integration through to a genuine Itô integral, the foundation earlier formalizations assume or sketch, and then consuming it. The companion paper [6] develops this in full; here is the spine.

The integral. The Itô integral $\mathcal{I}_T$ is a linear isometry from the square-integrable predictable integrands on $(0, T] \times \Omega$ into $L^2(\mu)$ (`itoIntegralCLM_T`), the unique continuous extension of the elementary integral on simple adapted processes, obtained by a π – λ density argument and Mathlib’s `LinearMap.extendOfNorm`. As a process it is the conditional projection of its terminal value, hence a continuous L^2 martingale, and it has a continuous modification that is a local martingale on the half-line. Its range together with the constants is all of $L^2(\mathcal{F}_T^B)$, which is martingale representation, and it satisfies the `BrownianMotion` package’s axiomatic characterisation of a stochastic integral. One worked example is the identity behind $\int_0^T B \, dB$: an integrand g_B , constructed as the limit of truncated step processes of B , has $\mathcal{I}_T(g_B) = \frac{1}{2}(B_T^2 - B_0^2 - T)$ (`itoIntegralCLM_T_brownian`). That statement does not itself name g_B as B ; the named form is an instance of Itô’s formula

under exponential growth, which does name its integrand, though that instance is not stated separately [6].

Itô's formula. In L^2 , for $f(t, x)$ whose partial derivatives up to f_{tt} , f_{tx} and f_{xxx} are bounded or grow at most like $e^{\lambda|x|}$, $f(T, B_T) - f(0, B_0) = \mathcal{I}_T(f_x(\cdot, B)) + \int_0^T (f_t + \frac{1}{2}f_{xx})(s, B_s) ds$ with the integrand named. For every C^3 function, with no growth condition, the compensated process is a continuous local martingale on the null-augmented filtration, by localization at exit times (ito_formula_unrestricted). For an Itô process $X = x_0 + \int b ds + \int \sigma dB$ with bounded adapted continuous coefficients and f with bounded f' and f'' , the formula holds at a fixed time (ito_formula_adapted).

The risk-neutral measure, derived. In most treatments, and in earlier formalized pricing, an equivalent martingale measure is posited and prices are its expectations. Here it is constructed. A static Girsanov change of measure by an Esscher density turns the physical Gaussian driver into one under which the terminal asset has exactly the lognormal law the Black–Scholes layer takes as its hypothesis, so that hypothesis is a theorem (BSCallHyp.exists_of_physical) and the call formula holds under a measure constructed from the physical Gaussian driver alone (bs_call_formula_of_physical). The dynamic Girsanov theorem is proved along a ladder of drifts (constant, simple adapted, continuous adapted, bounded predictable): under $\frac{dQ}{d\mu} = \exp(-\mathcal{I}_T(\theta) - \frac{1}{2} \int_0^T \theta^2 ds)$ the process $B + \int \theta ds$ starts at zero and has independent $N(0, t - s)$ increments on $[0, T]$, which fixes its finite-dimensional laws; path continuity and independence from the past are not stated. The discounted geometric Brownian motion is an Itô integral with its integrand named (discountedGBM_eq_itoIntegral), so the pricing layer meets the continuous world through the integral itself, not only through the marginal laws that the third version described.

Completeness and its limits. Every square-integrable $\mathcal{F}_T^B$ -measurable claim is $\mathbb{E}[H] + \int_0^T \varphi dB$ for a unique hedge (exists_replicating_strategy); integrating against the price $S = S_0 + \int \sigma dB$ instead, with $\sigma \neq 0$ almost everywhere, the hedge is a holding in the price. A probability measure with square-integrable density under which S is a martingale agrees with μ on $\mathcal{F}_T^B$: the direction of the second fundamental theorem from completeness to uniqueness. The converse, the general Girsanov theorem under Novikov's condition, and the identification of the hedge (Clark–Ocone) are open.

Equations and jumps. Lipschitz stochastic differential equations have a unique solution in the predictable L^2 space whenever $TL_b + \sqrt{T}L_\sigma < 1$ (picardMap_exists_unique_fixedPoint), realized as a sample-path process (sde_pathwise_decomposition). Jump-diffusions enter through a hypothesis structure (JumpDiffusionProcess): an adapted process with independent increments whose laws are those of drift plus Gaussian plus compound Poisson. On it the discounted price is a martingale exactly at the compensated drift, the conditional call price at every date is Merton's series [19], and two equivalent compensated laws at one date price some call differently. Such a process is constructed only without jumps, as Brownian motion with drift.

From lattice to continuum. The Cox–Ross–Rubinstein call price converges to the Black–Scholes price, by characteristic functions and Lévy's continuity theorem, with put–call parity removing the uniform-integrability obstruction. The limit is proved for call prices and for the law of the terminal log-return; Donsker's invariance principle is not formalized.

6 ARCHITECTURE: PRINCIPLES AND BRIDGES

The library stays coherent by deriving results from a few reusable principles rather than re-proving each instrument. Four pillars carry the mathematics, and a small set of bridges connects them.

I. No-arbitrage as convex duality. The finite fundamental theorem is a separating hyperplane between the attainable gains and the simplex; the Artzner–Delbaen–Eber–Heath representation of coherent risk measures [1] separates a rejected position from the acceptance cone. The two separations are sibling lemmas in one module, proved from the same two helper lemmas [5]. On a finite state space, pricing as a non-negative linear functional on payoffs gives put–call parity, the forward price and convexity of the call in the strike as corollaries of one structure rather than instrument-by-instrument results.

II. Stochastic calculus. Continuous-time models are built on Brownian motion, and the Itô integral and formula of Section 5 are the calculus for their functionals; pricing hypotheses stated on marginal laws are tied to Brownian motion by the bridges below.

III. The probabilistic–analytic duality. A price is both a risk-neutral expectation and the solution of a partial differential equation, and Feynman–Kac is the bridge. The Black–Scholes equation is derived by writing the price as a heat-kernel integral and using the heat equation for the kernel, without differentiating the closed form.

IV. Exponential families and intensities. Closed forms come from Gaussian, lognormal and exponential structure. One Esscher module serves three users: its tilt, Mathlib’s `Measure.tilted` at a linear exponent, gives the static Girsanov change of measure and Esscher pricing of jump-diffusions, and its moment-generating-function uniqueness lemma identifies Brownian increments under a new measure. And one exponential of an integrated intensity, `survivalFromIntensity`, is the shared definition of the intensity-based actuarial and credit survival functions.

The Garman normal form. Many Black–Scholes-family prices have the form $A \Phi(d_1) - K \cdot DF \cdot \Phi(d_2)$. The library proves this equality for the standard call, Black-76 futures options, payer swaptions, the call with a continuous dividend yield, and Margrabe’s exchange option [15]. The third version listed the put, digital and power options among the proved instances; they are not. Garman–Kohlhagen is an instance only through the dividend-yield case, without a theorem of its own.

Grounding in Brownian motion. Pricing hypotheses are stated marginally (`BSCallHyp` asks only that the terminal asset be lognormal under the pricing measure), and bridge modules show that they follow from a pre-Brownian motion, through which the flagship prices connect to the continuous-time object.

Certified unification. What a formal base over classical mathematics produces is not new theory but machine-checked identities between objects developed for different purposes. For Gaussian portfolios with a common mean, minimising value-at-risk or expected shortfall at a level above one half is equivalent to minimising variance (`isMinOn_valueAtRisk_iff_isMinOn_portfolioVarN`). The N -asset portfolio variance with uncorrelated assets of common variance is that variance times the Herfindahl concentration index (`portfolioVarN_diag_eq_herfindahl`), and with equal correlation it is an explicit mixture of the index and the squared total weight. Mortality and reduced-form credit share one intensity-based survival function. The third version presented this last identity as one between independently developed definitions; in fact both are defined through a

shared `survivalFromIntensity`, so it holds by definition, and it is the shared definition, not the identity, that records the unification.

A representation layer. The contract layer (`Contracts/`) is not a fifth principle but a representation across the four: payoffs become inductive objects, proved measurable and adapted to their own observation times, and for the European call, put, digital and capped call under single-asset Black–Scholes their values reduce to the closed forms the pricing layer already proves. Its design follows Bilokon’s *The Contract Is Not the Model* [2]; no code is taken from it.

7 THE FAITHFULNESS AUDIT

Formalization papers often leave implicit the gap between “we proved X ” and “we proved a weaker X under added hypotheses”. The library makes the gap explicit and partly machine-checkable.

7.1 TIERS AND DISCLOSURES

Every benchmark entry declares one of four statuses. `full`: the Lean statement is the mathematical claim, derived from foundational primitives. `library_wrapper`: a thin restatement of an upstream Mathlib or `BrownianMotion` theorem. `reduced_core`: an honest but narrower statement; in twelve of the thirteen current cases the textbook conclusion is a field of a structure and the theorem reads it off by projection, so nothing is derived, and in the thirteenth a special case is proved. `placeholder`: none remain. Delivery claims count only `full` and `library_wrapper`. (The third version glossed the reduced core as “holds under an added hypothesis or with an axiomatized sub-step”, which understated how little the twelve specifications prove.) Each entry’s description states what the entry proves, not the textbook theorem it is named after, and its `formalization_scope` gives the long-form disclosure of what it leaves out. Both are published, together with the Lean statement, in a public dataset [14].

7.2 MACHINE-CHECKED GATES

The tiers are metadata, checked by test gates that continuous integration runs before the Lean build, and the Lean build itself checks axioms.

- *Axiom audits.* A curated file pins the `#print axioms` output of 372 headline constants under `#guard_msgs`. A generated file pins 600: every library constant cited in proof position by any benchmark snippet (571), resolved by declaration through `opens`, dot notation and foreign namespaces, and the 29 upstream constants that wrapper entries re-export. Together they pin 760 distinct constants. The upstream pins matter because `BrownianMotion` contains `sorry`s at the pin; none lies beneath a pinned constant. Any axiom beyond `propext`, `Classical.choice` and `Quot.sound`, any `sorryAx`, or a renamed constant fails the build.
- *Text gates.* After comment stripping, library sources contain no `sorry`, `admit`, `native_decide`, `polyrith`, search-tactic suggestions or premise-selection calls.
- *Shape gates.* A full entry may not bundle an assumed `Prop`-valued structure (the reduced-core pattern), nor rest on a definitional `rfl` proof; every reduced-core description must say what is not derived; and an existential statement whose witness is never pinned down may not be described by prose that names an integral of it.

Benchmark entries	535
full	504
library_wrapper	18
reduced_core	13
placeholder	0
Delivery-ready (full + library_wrapper)	522
Lean files / lines under MathFin/ (excluding audit and blueprint files)	560 / 100,315
of which contributed ports (Section 9)	214 / 24,238
Theorem and lemma declarations	3,841
Constants pinned by the axiom audits (curated / generated / distinct)	372 / 600 / 760
Verification-ledger rows (all fresh)	535

TABLE 4. The library at release v1.5.0.

- *Evidence freshness.* A verification ledger records, for every entry, a hash of its snippet, the transitive library modules it imports and the toolchain pins under which it last verified; an entry whose inputs change becomes stale and must re-verify.

7.3 WHAT THE GATES CANNOT SEE

Two failure modes pass every mechanical gate. The first is vacuity. The Cox–Ross–Rubinstein convergence theorems that the third version of this paper reported as checked end to end carried the hypothesis that the n -step tree is arbitrage-free for every n , which fails at $n = 0$, where the step size is $T/0 = 0$; so the theorems were true of no market and said nothing. No axiom audit, kernel replay or sorry gate can detect this. It was found by reading the statement, not by any gate, and fixed by quantifying over $n \geq 1$ and proving that the hypothesis then holds whenever $|r|\sqrt{T} < \sigma$ (`binomialNoArb_crr`). The external check of Section 7.4 likewise pairs its theorem with a satisfiability witness.

The second is prose that outruns its statement: the build is green precisely because the statement is weaker than the description. The library has repeatedly found such drift in its own text, including the Girsanov results, once described as producing a Brownian motion when they fix finite-dimensional laws, and the integral $\int B dB$ of Section 5. Only its mechanical slice is gated. The rest is a review discipline: every change to proof content closes with a multi-agent review against eight lenses, whose standing first pass reads every touched description, docstring and paragraph against its statement, and whose cadence (one review per twelve new entries at most) is itself test-enforced. This paper was revised with the same first pass; the corrections listed in the introduction are what it found.

7.4 EXTERNAL CHECKS

For the coherent-risk representation theorem the repository carries a Mathlib-only re-statement, with a witness that coherent risk measures exist, and a proof of both from the library that every build re-checks; the external checker Comparator [7] is configured to confirm that the proved statements are exactly the restated ones and that only the three standard axioms are used. A kernel replay of every declaration through `leanchecker` exists as a workflow but has never completed: the full Mathlib environment it loads does not fit in the memory of a hosted runner.

8 EVALUATION

Table 4 gives the numbers, from the repository’s coverage report and source tree. The reduced cores sit at the genuinely hard frontier of continuous-time and discrete-process theory: the reflection principle, nowhere differentiability and the law of the iterated logarithm for Brownian motion; Novikov’s condition, the general Girsanov theorem, Lévy’s characterisation and the two-dimensional Itô formula; five theorems on Markov chains (the strong Markov property, recurrence criteria, uniqueness of the stationary distribution, the ergodic theorem and convergence to equilibrium); and, as a proved special case, the exponential law of the first interarrival time of a Poisson process. The mathematical-finance benchmark, the largest, has 336 of its 337 entries full. The library is most faithful where the mathematics is classical and elementary, and its partial cores sit where the infrastructure is missing.

Reproducibility. The toolchain is pinned: Lean v4.33.0-rc1 [9], Mathlib at 0434c03 and BrownianMotion at 0d5b6eb, in `lean-toolchain` and `lake-manifest.json`. A plain lake build from the repository root checks every proof; release v1.5.0 is archived on Zenodo [26], and a pinned container image (`ghcr.io/formal-applied-math/mathfin-verify`) reproduces the build.

9 PROVENANCE

A library that audits its claims should also account for its code. The repository records provenance file by file and in a machine-readable self-report.

Contributed developments. About a quarter of the source lines are ports of two developments by Robert Martin, each kept under its original authorship. The first proves, in the constant-parameter Black–Scholes model with $r > 0$ and dividend yield $0 \leq q \leq r$, that the American put’s early-exercise boundary $B(\tau)$ is strictly convex in the time to expiry and that $\log(B(\tau)/K)$ is convex [16]. The value is the supremum over all stopping times of the completed Brownian filtration, and the headline theorems assume no classical solution of the free-boundary problem; a further, conditional part of the port, stated over an assumed classical solution that nothing shows inhabited, is not claimed. The second gives a fixed Boolean combination of signs of polynomials in the five raw-SVI parameters that holds exactly when the parameters are admissible and Durrleman’s condition holds, that is, when the call priced through the SVI smile is convex in the strike [17, 12]. The library adds the large-strike tail condition: with one further sign the formula holds exactly when that call is a normalised call price function in Roper’s sense, which is freedom from butterfly arbitrage [22]; recovering the risk-neutral law from such a function is not formalized. The repository’s survey of open problems had listed both questions as open, the first for $0 < q < r$ and the second as an explicit semialgebraic description of the SVI domain, and records both as resolved by these developments. The credit is Robert Martin’s; the first was, on its author’s report, substantially AI-assisted; neither has been refereed; and their status is that of machine-checked, axiom-pinned results.

Other sources. Nagy’s derivation [20] and the automated-market-maker formalization of Pusceddu and Bartoletti [21] are adapted in four modules with attribution headers. The contract layer takes design ideas, but no code, from Bilokon [2]. Yosuke Ito’s Isabelle entry on actuarial mathematics was consulted, with permission, for the survival layer. Alfredo Garcia contributed the Glosten–Milgrom model and the drifted price path, and Aleksey Salkutsan downside performance metrics and forward-rate agreements. Much of

the benchmark follows Saporito’s lecture notes on stochastic processes [23], the corpus spine.

AI assistance. The library was written with AI assistance, and the repository says so. The author used an AI coding assistant (Claude Code) as an authoring assistant throughout. Four benchmark entries came from an automated pipeline in which language models drafted and proved the statements; each was reviewed and refined by hand, and the pipeline is treated as a scout, not an author. The gates of [Section 7](#) apply to all code unchanged, whoever or whatever wrote it, and every proof is checked by the Lean kernel when the library builds. This manuscript was revised with the same assistant, and its statements were checked against the Lean sources of release v1.5.0.

10 LIMITATIONS

The limitations of the headline results are those their statements carry. Girsanov’s theorem is proved for bounded predictable drifts and at the level of finite-dimensional laws. Of the second fundamental theorem only the direction from completeness to uniqueness is proved, and martingale representation proves that a hedge exists and is unique without identifying it. Itô’s formula for an Itô process is proved at a fixed time for bounded adapted continuous coefficients and C^2 functions f with bounded first and second derivatives. Stochastic differential equations are solved only on horizons with $TL_b + \sqrt{T}L_\sigma < 1$. The jump-diffusion results assume the process (`JumpDiffusionProcess`). The multi-period fundamental theorem, for one risky asset, and the representation of coherent risk measures are proved on finite probability spaces; the general multi-period theorem of Dalang, Morton and Willinger is open. Several results are narrower than their names: the Margrabe grounding asserts the existence of its pricing measure and driver, the Vasicek integrated rate is defined through its Wiener representation, and the market-making result solves only the approximate Hamilton–Jacobi equation.

The apparatus has limits of its own. Library theorems that no benchmark entry cites and the curated audit omits are not axiom-pinned; the text gate and the Lean kernel still apply to them. Continuous integration checks that the ledger is fresh, not that it was re-verified; re-verification runs on demand. The public dataset publishes descriptions and scope disclosures but not the provenance notes. And the kernel replay has never completed.

11 CONCLUSION

The library is, to our knowledge, the most comprehensive machine-checked development of mathematical finance to date, but its real contributions are not its size. They are its depth, a constructed Itô integral on which the objects of continuous-time pricing are derived, and an audit that makes every claim’s logical status checkable and keeps prose from outrunning statements.

What the library is *not* is equally part of the point. Machine-checking classical financial mathematics yields certified unification of known results, not new financial theory, and the tiers, disclosures and gates make that honesty checkable. A formal base can nonetheless host new results once they exist: two contributed developments address questions its own survey had listed as open, and the library’s apparatus is what lets a reader see exactly what they prove and on what they rest. The open frontier is the natural next layer for a community library: the general multi-period Dalang–Morton–Willinger theorem and its continuous-time counterpart, Girsanov’s theorem under Novikov’s condition, Itô’s

formula for Itô processes without bounds, pathwise quadratic variation, the Clark–Ocone formula, and the thirteen results still classified as reduced cores.

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